National Research University Higher School of Economics (HSE)

Basic Curriculum

Field of study 01.04.02 Applied Mathematics and Informatics
Educational Programme "Math of Machine Learning"
Trajectories: "Math of Machine Learning"
Implementing unit: Faculty of Computer Science, HSE - Moscow

APPROVED
09.04.2026
Vice Rector
ROSHCHIN S.Y.
Signed with EDS

Years of Study: 2026/2027 - 2027/2028
Length of Programme: 2 years
Mode of Study: Full Time
Degree: Master’s degree / MBA

Block Code	Course	Subject type	Credits	Credits by Years		Planned Educational Programme Development Results
				1	2	
	Degree Programme		120,00	60,00	60,00	
	Math of Machine Learning (General track)		120,00	60,00	60,00	
	Key Seminars		15,00	6,00	9,00	
1	Mentor's Seminar "Math of Machine Learning " (offered in a foreign language)	C	15,00	6,00	9,00	GPC-3.AMI, GPC-4.AMI, PC-1, UC-2, UC-3, UC-4
	Internship		30,00	6,00	24,00	
	Preparation of the Graduate Thesis		24,00		24,00	
1	Graduation Thesis (offered in a foreign language)	C	24,00		24,00	GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, GPC-4.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6
	Fixed		6,00	6,00		
1	Innovation Workshop (offered in a foreign language)	C	6,00	6,00		GPC-2.AMI, GPC-4.AMI, PC-1, PC-3, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6
	Major		69,00	48,00	21,00	
	Basic Components		30,00	27,00	3,00	
1	Additional Topics in Probability Theory (offered in a foreign language)	C	3,00	3,00		GPC-1.AMI, PC-6, UC-6
2	Markov Chains (offered in a foreign language)	C	3,00	3,00		GPC-1.AMI, PC-1, PC-2, UC-6
3	Sampling and Generative Modeling (offered in a foreign language)	C	3,00	3,00		GPC-2.AMI, GPC-3.AMI, PC-1, PC-3, UC-1, UC-2, UC-3, UC-4
4	High Dimensional Probability and Statistics (offered in a foreign language)	C	6,00	6,00		GPC-2.AMI, PC-3, UC-6

5	Machine Learning (offered in a foreign language)	C	6,00	6,00		GPC-2.AMI, GPC-3.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, UC-6
6	Computational Linear Algebra (offered in a foreign language)	C	6,00	6,00		GPC-1.AMI, PC-3, UC-6
7	Modern Algorithmical Optimization (offered in a foreign language)	C	3,00		3,00	GPC-2.AMI, PC-1, PC-4, UC-1
	Elective Courses. 1 year		21,00	21,00		
1	Introduction to Data Analysis (offered in a foreign language)	E	3,00	3,00		GPC-3.AMI, PC-4, UC-2
2	Computational Imaging (offered in a foreign language)	E	3,00	3,00		GPC-2.AMI, PC-1, PC-4, PC-6, UC-2
3	Foundations of Software Engineering (offered in a foreign language)	E	3,00	3,00		GPC-4.AMI, PC-5, UC-2
4	Principles of Applied Statistics (offered in a foreign language)	E	3,00	3,00		GPC-1.AMI, PC-3, UC-1
5	Introduction to Computer Vision (offered in a foreign language)	E	3,00	3,00		GPC-2.AMI, PC-2, PC-4, UC-3
6	Optimization in Machine Learning (offered in a foreign language)	E	3,00	3,00		GPC-2.AMI, PC-1, UC-1
7	Perception in Robotics (offered in a foreign language)	E	6,00	6,00		GPC-2.AMI, PC-3, PC-4, UC-2
8	Tensor Decompositions and Tensor Networks in Artificial Intelligence (offered in a foreign language)	E	6,00	6,00		GPC-2.AMI, PC-1, PC-3, UC-1
9	Deep Learning (offered in a foreign language)	E	6,00	6,00		GPC-4.AMI, PC-1, PC-4, UC-2, UC-6
10	Geometrical Methods of Machine Learning (offered in a foreign language)	E	3,00	3,00		GPC-2.AMI, PC-1, UC-1
11	Deep Learning for Natural Language Processing (offered in a foreign language)	E	3,00	3,00		GPC-1.AMI, PC-1, PC-2, PC-3, UC-2
	Elective Courses. 2 year		18,00		18,00	
1	Mathematical Foundations of Reinforcement learning (offered in a foreign language)	E	3,00		3,00	GPC-1.AMI, GPC-2.AMI, PC-3, PC-6, UC-6
2	Optimal Transport and Its Applications (offered in a foreign language)	E	3,00		3,00	GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, PC-1, PC-3, PC-4, UC-1
3	Leadership for Innovators (offered in a foreign language)	E	3,00		3,00	GPC-4.AMI, PC-6, UC-1, UC-3, UC-4, UC-6
4	Generative Models in Machine Learning (offered in a foreign language)	E	3,00		3,00	GPC-4.AMI, PC-5, PC-6, UC-2, UC-6
5	Reinforcement Learning (offered in a foreign language)	E	6,00		6,00	GPC-2.AMI, PC-1, PC-3, UC-2
6	High Performance Python Lab (offered in a foreign language)	E	3,00		3,00	GPC-3.AMI, PC-3, PC-5, UC-1
7	Planning Algorithms in Artificial Intelligence (offered in a foreign language)	E	3,00		3,00	GPC-2.AMI, PC-1, PC-3, PC-6, UC-1

8	Theoretical Methods of Deep Learning (offered in a foreign language)	E	3,00		3,00	GPC-1.AMI, GPC-2.AMI, UC-1, UC-2
	Final State Certification (FSC)		6,00		6,00	
1	Final Qualification Paper (offered in a foreign language)	C	6,00		6,00	GPC-1.AMI, GPC-2.AMI, GPC-3.AMI, GPC-4.AMI, PC-1, PC-2, PC-3, PC-4, PC-5, PC-6, UC-1, UC-2, UC-3, UC-4, UC-5, UC-6

Curriculum agreed:

Academic Supervisor	Naumov A.A.	08.04.2026
Dean	ARZHANTSEV I.V.	08.04.2026
Head of Centre for Educational Model Design	LEPESHKIN I.A.	08.04.2026

Образовательная программа направлена на формирование и развитие следующих компетенций (результатов освоения):

Competence code	Name of the competence
UC-1	Способен осуществлять критический анализ проблемных ситуаций на основе системного подхода, вырабатывать стратегию действий.
UC-2	Способен управлять проектом на всех этапах его жизненного цикла.
UC-3	Способен организовывать и руководить работой команды, вырабатывать командную стратегию для достижения поставленной цели.
UC-4	Способен применять современные коммуникативные технологии, в том числе на иностранном(ых) языке(ах), для академического и профессионального взаимодействия.
UC-5	Способен анализировать и учитывать разнообразие культур в процессе межкультурного взаимодействия.
UC-6	Способен определять и реализовывать приоритеты собственной деятельности и способы ее совершенствования на основе самооценки и образования в течении всей жизни.
GPC-1	Способен решать актуальные задачи фундаментальной и прикладной математики.
GPC-2	Способен совершенствовать и реализовывать новые математические методы решения прикладных задач.

GPC-3	Способен разрабатывать математические модели и проводить их анализ при решении задач в области профессиональной деятельности.
GPC-4	Способен комбинировать и адаптировать существующие информационнокоммуникационные технологии для решения задач в области профессиональной деятельности с учетом требований информационной безопасности.
PC-1	Able to find relevant areas of scientific research, set up and test scientific hypotheses
PC-2	Able to analyze product tasks and propose solutions based on data
PC-3	Able to build hypotheses and test them based on data
PC-4	Able to choose according to the task being solved and build machine learning models, as well as evaluate their quality
PC-5	Able to understand the architecture of large projects and implement new functionality in them
PC-6	Able to apply in practice the acquired fundamental and applied knowledge for the selection of an actual task and the organization of the development process

* Subject type:

Compulsory course	C
Elective course	E